NMEC595 - Research Methodology

Lecture 12: Confidence Intervals

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Outline

Lesson Objectives

What You Will Learn:

Upon completion of this lesson, you should be able to:

1. Describe the role of **statistical inference** in estimation
2. Explain the **general form** of a confidence interval
3. **Construct** confidence intervals for population mean or proportion
4. **Interpret** confidence intervals correctly
5. Identify when to use t-**distribution** vs normal distribution
6. Define and interpret **margin of error**
7. Calculate **required sample size** for desired margin of error

Statistical Inference

Definition:
The real power of statistics comes from applying probability concepts to situations where you have **data** but not necessarily the **whole population**.

Two Types of Statistical Inference:

5.1 Estimation
Use sample information to **estimate** the parameter of interest.

Example: Poll results to estimate minister's approval rating

5.2 Hypothesis Tests
Use sample information to **test** whether a statement about the parameter is true.

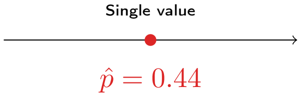
Example: Test if claimed approval rating of more than 75% is supported by the poll data

Types of Estimates

1. Point Estimate
An estimate that is **one numerical value**.

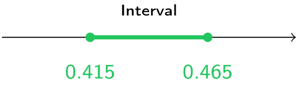
Examples:

- Sample mean $\bar{x}$
- Sample proportion $\hat{p}$



2. Interval Estimate Gives an interval as the estimate for a parameter.

- Built around point estimates
- Quantifies uncertainty



2a. Confidence Interval
An interval of values computed from sample data that is **likely to cover the true parameter** of interest.

General Structure

General Form of a Confidence Interval

$$\text{Sample Statistic} \pm \text{Margin of Error}$$

Margin of Error

$$\text{Margin of Error} = M \times \hat{SE}(\text{estimate})$$

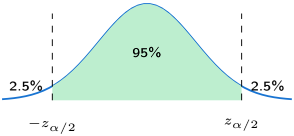
where M is a multiplier that depends on the confidence level

Example: President's Approval Rating

- Sample proportion: 44%
- Margin of error: $\pm 3.5\%$
- Confidence interval: **40.5% to 47.5%**

Components

1. **Point Estimate (Center)**
   - Best single guess
   - $\bar{x}$ for mean
   - $\hat{p}$ for proportion
2. **Standard Error**
   - Measures uncertainty
   - Usually estimated from sample
   - $\sqrt{\frac{\hat{p}(1-\hat{p})}{n}}$ for proportion
   - $\frac{s}{\sqrt{n}}$ for mean
3. **Multiplier (M)**
   - Calculated from the same distribution as the sampling distribution of the point estimate.
   - Depends on confidence level
   - z for proportion, t for mean



Interpretation:

We are $(1-\alpha)100\%$ confident that the parameter is in the interval.

Confidence Level

Why not be 100% confident?

The 100% Confidence Problem

"I am 100% confident the president's approval rating is between 0% and 100%."

This is useless! It provides no information.

Common Confidence Levels:

Confidence Level	α	Usage
90%	0.10	Less common
95%	0.05	**Most widely used**
99%	0.01	Very conservative

Important Note

When you see a margin of error without confidence level stated, assume **95% confidence**.

Part-A: Point Estimate for Proportion

Point Estimate

$$\hat{p} = \frac{\text{\# of successes in sample}}{\text{sample size } n}$$

Recall from Lesson 4:

If $np \geq 5$ and $n(1-p) \geq 5$, then $\hat{p}$ is approximately Normal with:

- Mean: p
- Standard error: $\sqrt{\frac{p(1-p)}{n}}$

Problem

We don't know p! So we **estimate** the standard error using $\hat{p}$:

$$\hat{SE}(\hat{p}) = \sqrt{\frac{\hat{p}(1-\hat{p})}{n}}$$

Three Steps to Construct CI

Step 1: Check Conditions

Since p is unknown, check:

- $n\hat{p} > 5$ and
- $n(1-\hat{p}) > 5$

Step 2: Construct the General Form

$(1-\alpha)100\%$ CI for Population Proportion

$$\hat{p} \pm z_{\alpha/2}\sqrt{\frac{\hat{p}(1-\hat{p})}{n}}$$

where $z_{\alpha/2}$ is the z-value with area $\alpha/2$ to the right

Step 3: Interpret the CI

We are $(1-\alpha)100\%$ confident that the population proportion is between [lower bound] and [upper bound].

Derivation of the Confidence Interval

To calculate the confidence interval, we need to know how to find the z-multiplier. So where does this z_α come from?

$$P\left(-z_{\alpha/2} \le \frac{\hat{p}-p}{\sqrt{\frac{\hat{p}(1-\hat{p})}{n}}} \le z_{\alpha/2}\right) = 1-\alpha$$

Rearranging gives

$$P\left(\hat{p} - z_{\alpha/2}\sqrt{\frac{\hat{p}(1-\hat{p})}{n}} \le p \le \hat{p} + z_{\alpha/2}\sqrt{\frac{\hat{p}(1-\hat{p})}{n}}\right) = 1-\alpha$$

The figure shows the general confidence interval on the normal curve.

Finding the Z-Multiplier

Example 5.1: Find $z_{0.15}$

$z_{0.15}$ means $P(Z > z_{0.15}) = 0.15$
This implies $P(Z \leq z_{0.15}) = 0.85$
From standard normal table: $z_{0.15} = 1.04$

Common Z-Multipliers:

Confidence	α	$z_{\alpha/2}$	**Multiplier**
90%	0.10	$z_{0.05}$	1.645
95%	0.05	$z_{0.025}$	**1.960**
98%	0.02	$z_{0.01}$	2.326
99%	0.01	$z_{0.005}$	2.576

Observation

Higher confidence $\Rightarrow$ Larger multiplier $\Rightarrow$ Wider interval

Example

Problem

A random sample of 1500 U.S. adults is taken. Of the 1500 surveyed, 660 approved of the president's performance. Calculate a 95% confidence interval for the overall approval rating.

Step 1: Check Conditions

- Successes: $660 \geq 5$ ✓
- Failures: $840 \geq 5$ ✓

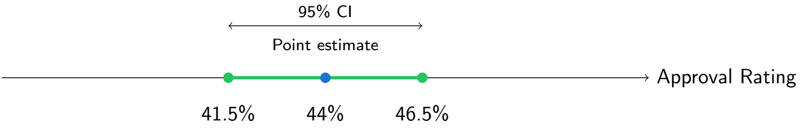
Step 2: Construct CI

$\hat{p} = \frac{660}{1500} = 0.44, \quad n = 1500, \quad z_{0.025} = 1.96$

$$0.44 \pm 1.96\sqrt{\frac{0.44(1-0.44)}{1500}}$$

$$= 0.44 \pm 0.025$$

Example



Interpretation

"We are **95% confident** that the overall U.S. adult approval rating for the current president is from **41.5% to 46.5%**."

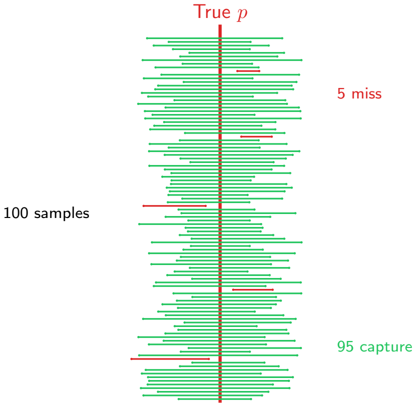
Alternative Statement:

"The current U.S. approval rating for the president is 44% with a 95% margin of error of 2.5%."

Important

Use **proportion** (0.415, 0.465) or **percentage** (41.5%, 46.5%) – be consistent and clear!

What Does 95% Confidence Mean?



Correct Interpretation

If we repeatedly draw samples and calculate 95% CIs, about **95% of the intervals** will contain the true parameter p.

Sample Size for Proportion CI

What should be the sample size?

Given desired margin of error E and confidence level, we need:

Method 1: Educated Guess

$$n = \frac{z_{\alpha/2}^2 \hat{p}_g (1 - \hat{p}_g)}{E^2}$$

where $\hat{p}_g$ is an educated guess for p. Used if it is cheap to sample more elements when needed.

Method 2: Conservative Method

$$n = \frac{z_{\alpha/2}^2 (0.5)^2}{E^2}$$

Uses $p = 0.5$ (worst case, maximum variability). Used if start-up cost of sampling is expensive and thus not economical to sample more elements later.

Always round UP to next integer!

Example

Problem

For the next poll of president's approval rating, we want margin of error of 1% with 95% confidence. Previous poll showed 72% approval. How many individuals should we sample?

Educated Guess Method:

$z_{0.025} = 1.96$, $E = 0.01$, $\hat{p}_g = 0.72$

$$n = \frac{(1.96)^2(0.72)(1-0.72)}{(0.01)^2} = 7744.67 \Rightarrow \mathbf{7745}$$

Conservative Method:

$$n = \frac{(1.96)^2(0.5)(1-0.5)}{(0.01)^2} = 9604 \Rightarrow \mathbf{9604}$$

Part-B: Point Estimate and Interval Estimate for μ

Point Estimate

$$\bar{x} = \text{sample mean}$$

Goal: use $\bar{x}$ to estimate the population mean μ, and quantify uncertainty with a confidence interval.

If σ were known

$$\bar{x} \pm z_{\alpha/2} \frac{\sigma}{\sqrt{n}}$$

In practice, however, the population standard deviation σ is usually **unknown**.

The t-Distribution

If the population standard deviation σ were known, then

$$\frac{\bar{x} - \mu}{\sigma/\sqrt{n}} \sim N(0,1).$$

But in practice, we usually do **not** know σ.
So we replace σ by the sample standard deviation s:

$$\frac{\bar{x} - \mu}{s/\sqrt{n}}.$$

Key Idea

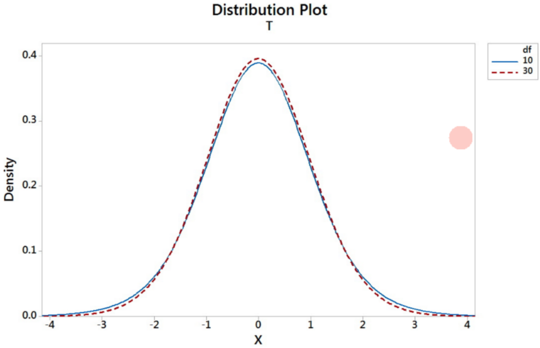
Replacing σ by s introduces extra variability. Because of this extra uncertainty, the statistic no longer follows a standard normal distribution.

Instead, when the population is Normal,

$$\frac{\bar{x} - \mu}{s/\sqrt{n}} \sim t_{n-1}.$$

The t-Distribution

Discovered by William Sealy Gosset (pen name: Student) in 1908



$$\frac{\Gamma\left(\frac{\nu+1}{2}\right)}{\sqrt{\pi\nu}\,\Gamma\left(\frac{\nu}{2}\right)}\left(1+\frac{x^2}{\nu}\right)^{-\frac{\nu+1}{2}}$$

- ν denotes the degrees of freedom (df)

The t-Distribution

Properties:

1. Symmetric about 0
2. Has heavier tails than the Normal distribution
3. Depends on the degrees of freedom, $df = n - 1$ (why?)

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Properties:

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   When we replace σ by the sample standard deviation s, we use the same data to estimate the center and the spread. Since one piece of freedom is already used up in estimating μ, only $n - 1$ independent pieces of information remain for estimating variability.

4. Approaches the Normal distribution as n becomes large
5. t_α is the t-value with area α to the right

Confidence Interval for Mean (Unknown σ)

Conditions: at least one of the following should hold:

- The population distribution is Normal, or
- The sample size is large ($n \geq 30$)

$(1-\alpha)100\%$ CI for μ

$$\bar{x} \pm t_{\alpha/2,\, n-1} \frac{s}{\sqrt{n}}$$

Interpretation: We are $(1-\alpha)100\%$ confident that the population mean μ lies between

$$\bar{x} - t_{\alpha/2,\, n-1} \frac{s}{\sqrt{n}} \quad \text{and} \quad \bar{x} + t_{\alpha/2,\, n-1} \frac{s}{\sqrt{n}}.$$

Example

Problem

A random sample of 50 patients who visited the ER over the past week showed mean wait time of 30 minutes and standard deviation of 20 minutes. Find a 95% confidence interval for the average ER wait time.

Step 1: Check Conditions

$n = 50 \geq 30$ ✓ (Large sample, can use t-interval)

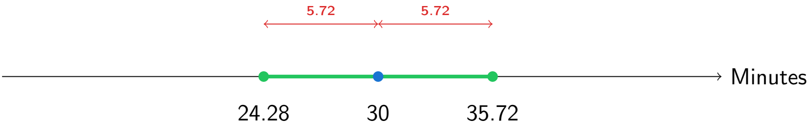
Step 2: Construct CI

$\bar{x} = 30$, $s = 20$, $n = 50$, $df = 49$

For 95% CI: $t_{0.025,49} \approx 2.021$ (from table with $df = 40$)

$$30 \pm 2.021 \frac{20}{\sqrt{50}}$$
$$= 30 \pm 5.72$$

Example



Interpretation

We are **95% confident** that the mean emergency room wait time at our local hospital is from **24.28 minutes to 35.72 minutes**.

R Output (using exact $df = 49$):

N	Mean	StDev	SE Mean	95% CI for μ
50	30.00	20.00	2.83	(24.32, 35.68)

Slight difference due to using $t_{0.025,49} = 2.010$ (more precise)

Small Sample: Checking Normality

What if $n < 30$?

Requirement

For small samples, we must check if the data come from a **Normal distribution**.

Tool: Normal Probability Plot

- Plots sample data against theoretical normal quantiles
- If points fall approximately on a straight line $\Rightarrow$ Normal distribution reasonable
- If points deviate significantly $\Rightarrow$ Normality questionable

Decision Rule

If all points fall within confidence bands on normal probability plot, it is **reasonable** to assume normality.

Example

Problem

12 rattlesnakes from central Pennsylvania measured (inches):

40.2, 43.1, 45.5, 44.5, 39.5, 38.5, 40.2, 41.0, 41.6, 43.1, 44.9, 42.8

Find a 90% confidence interval for mean length.

Step 1: Check Conditions

$n = 12 < 30$, so we need to check normality using normal probability plot. Normal probability plot shows all points within confidence bands ⇒ Normality reasonable ✓

Step 2: R Output

N	Mean	StDev	SE Mean	90% CI for μ
12	42.075	2.257	0.652	(40.905, 43.245)

Step 3: Interpretation

We are 90% confident that the population mean length of rattlesnakes is between **40.905 and 43.245 inches**.

Sample Size for Mean CI

How many samples do we need?

Crude Method (Conservative)

$$n = \left(\frac{z_{\alpha/2}\sigma}{E}\right)^2$$

where:

- E = desired margin of error
- σ can be estimated by $\frac{\text{range}}{4}$

Always round UP to next integer.

Sample Size for Mean CI

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where:

- E = desired margin of error
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Always round UP to next integer.

Example problem:
Estimate average holiday spending within \$120 with 90% confidence. Spending ranges from \$100 to \$1700.

$\sigma \approx \frac{1700-100}{4} = 400, \quad z_{0.05} = 1.645, \quad E = 120$

$$n = \left(\frac{1.645 \times 400}{120}\right)^2 = 30.07 \Rightarrow \textbf{31 students}$$

Estimating n with iterative method

Key Idea

The t-value depends on n (via degrees of freedom). Therefore, **iteratively** evaluate:

$$n = \left(\frac{t_{\alpha/2}\, s}{E}\right)^2$$

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Example $(s = 400, \ E = 120, \ \alpha = 0.05)$:

Iteration 1 — start with $n = 20$

df $= 19 \implies t_{0.05} = 1.729$

$$n = \left(\frac{1.729 \times 400}{120}\right)^2 = 33.21$$

$\Rightarrow$ use $n = 34$

Iteration 2 — try $n = 34$

df $= 33 \implies t_{0.05} = 1.697$

$$n = \left(\frac{1.697 \times 400}{120}\right)^2 = 31.99$$

$\Rightarrow$ use $n = 32$

Convergence

Plugging $n = 32$ reproduces the same result. $\therefore$ **Required sample size = 32 students.**

CI Formulas: Quick Reference

Parameter	**Conditions**	$(1-\alpha)100\%$ **CI**
Proportion p	$n\hat{p} \geq 5$ $n(1-\hat{p}) \geq 5$	$\hat{p} \pm z_{\alpha/2}\sqrt{\frac{\hat{p}(1-\hat{p})}{n}}$
Mean μ (σ known)	Population Normal	$\bar{x} \pm z_{\alpha/2}\frac{\sigma}{\sqrt{n}}$
Mean μ (σ unknown)	Normal population OR $n \geq 30$	$\bar{x} \pm t_{\alpha/2}\frac{s}{\sqrt{n}}$ $(df = n-1)$

Key Concepts Review

1. **Three Steps Always:**
    - Check conditions
    - Construct CI
    - Interpret CI
2. **Interpretation Template:**
   "We are [confidence level] confident that the [parameter] is between [lower bound] and [upper bound]."
3. **Margin of Error Tradeoffs:**
    - Higher confidence $\Rightarrow$ Wider interval
    - Larger sample size $\Rightarrow$ Narrower interval
4. **When to Use What:**
    - Proportion: Use z-distribution
    - Mean (σ known): Use z-distribution
    - Mean (σ unknown): Use t-distribution

Common Mistakes to Avoid

1. Misinterpreting the CI

WRONG: "There is a 95% probability that μ is in (24.28, 35.72)"
RIGHT: "We are 95% confident that μ is in (24.28, 35.72)"

2. Forgetting to Check Conditions

Always verify before using formulas:

- For proportion: $n\hat{p} \geq 5$ and $n(1 - \hat{p}) \geq 5$
- For mean: Normality or $n \geq 30$

3. Rounding Sample Size Down

ALWAYS ROUND UP! If $n = 30.07$, use $n = 31$.

Looking Ahead

What We Learned

- How to **estimate** population parameters
- How to **quantify uncertainty** with confidence intervals
- How to **determine sample size** for desired precision

Next Lesson: Hypothesis Testing

The second type of statistical inference:

Instead of estimating parameters, we'll **test claims** about them.

Example: Is the president's approval rating **more than 75%**?

Confidence intervals and hypothesis tests are **closely related**!